

HACKATHON 2026

Graph-Powered Fraud Intelligence System

AI-Powered Transaction Monitoring & Financial Crime Detection System

Detecting the Invisible through **Graph Forensics** and **Explainable Intelligence**.

The Information Overload Challenge

Financial crime teams are drowning in alerts.



High False Positives

Traditional rule-based systems flag too many legitimate transactions, wasting valuable analyst time.



Complex Fraud Rings

Fraudsters don't act alone. Single-transaction analysis misses coordinated attacks across merchants.

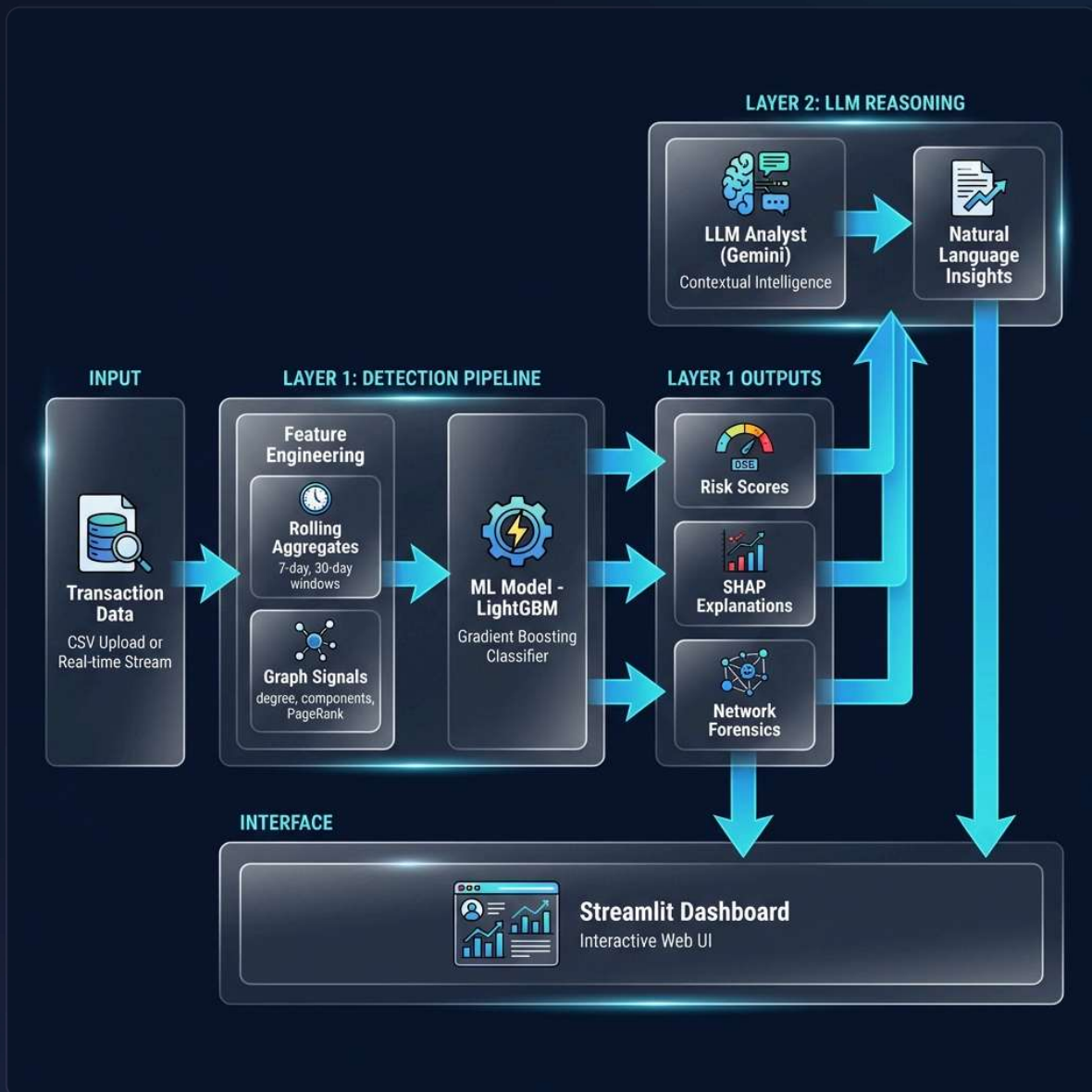


Black Box AI

"Computer says no" isn't enough. Analysts need to know *why* an alert was triggered to trust it.

Our Solution: Context-Aware Intelligence

We moved beyond simple rows of data to richer, connected insights.



METHODOLOGY

AI Engine & Algorithms

A robust pipeline built on Gradient Boosting and Graph Theory.



LightGBM Model

Selected for high efficiency on tabular data and ability to handle large datasets. Trained on **rolling aggregate features**.



Graph Algorithms

Connected Components & **Degree Centrality** are used to identify nodes acting as hubs in fraud rings.



Data Balancing

Handles class imbalance using `class_weight='balanced'` to ensure rare fraud cases are detected.

REQUIREMENTS

Technology Stack

Core Software

- **Python 3.9+**: Core logic
- **Streamlit**: Interactive Web UI
- **Pandas & NumPy**: Data Processing

AI & Vis

- **LightGBM**: ML Classifier
- **SHAP**: Explainability Engine
- **Graphviz**: Network Visualization
- **Altair**: Interactive Charts

Hardware

- **CPU**: Multi-core (Recommended for Training)
- **RAM**: 8GB+ (For In-Memory Graph Processing)
- **Storage**: 500MB+ for Dependencies & Models

INNOVATION 1

Network Forensics

Visualizing the connections that matter.

Graph Visualization

Instantly see if a card is connected to a "bad" merchant cluster.

Neighborhood Analysis

Zoom into specific transactions to see their immediate risk network (1-hop connections).

INNOVATION 2

Explainable Trust (SHAP)

Building trust between Human and Machine.

The "Why" Factor

Every alert comes with a SHAP breakdown. "Is it the amount? The location? The frequency?"

Analyst Empowerment

Analysts can validate the AI's logic instantly.

INNOVATION 3

LLM-Powered Contextual Intelligence

Adding a second reasoning layer for deeper insights.

Natural Language Analysis

LLM generates executive summaries, identifies critical patterns, and explains fraud in plain language.

Contextual Reasoning

Synthesizes LightGBM scores, SHAP features, and graph signals into actionable recommendations.

Investigation Assistant

Answers "Which alert should I prioritize?" and "What patterns am I seeing today?"

Actionable Insights

A premium interface designed for focus and clarity.

- **Descriptive Analytics** : Peak hours, risk hotspots, and amount distributions.
- **Dark Mode & Glassmorphism**: Reduced eye strain.
- **Seamless Workflow**: Upload → Analyze → Visualize → Review.

Business Impact



Speed

50% reduction in investigation time via visual graph tools.



Precision

Catching sophisticated fraud rings that rule-based systems miss.



Trust

Transparent AI ensures compliance and simplifies audits.

Thank You

Ready to detect the invisible.

[Replay Presentation](#)